

SUMMATIVE ASSESSMENT: EVALUATING LEARNING OUTCOMES IN THE FORM OF TERMINAL/FINAL EXAMINATION

Summative assessment is a fundamental component of the educational process, designed to evaluate student learning at the conclusion of an instructional period. Unlike formative assessment, which provides ongoing feedback during the learning process, summative assessment serves as a final measure of what students have learned. Typically administered at the end of a unit, course, or academic year. Summative assessment aims to determine the extent to which educational objectives have been achieved.

The primary purpose of summative assessment is to assess the overall effectiveness of instruction and learning. It provides a conclusive evaluation of student performance, often in the form of tests, final projects, or standardized exams. These assessments generate grades or scores that reflect a student's achievement in a given subject area over a specific period or time duration.

Summative assessment is often used to make critical decisions regarding student progression, certification, or placement in subsequent educational levels. Additionally, summative assessments provide valuable data that inform curriculum development and instructional strategies. By analyzing summative assessment results, educators can identify trends, strengths, and weaknesses within their instructional approaches, allowing for improvements in future teaching.

In conclusion, summative assessment plays a critical role in the educational process by providing a final evaluation of student learning. While it differs from formative assessment in its focus and application, it is an essential tool for measuring academic achievement. When balanced with formative assessments, summative assessments contribute to a well-rounded and effective approach to evaluating and supporting student learning.

National Curriculum of Pakistan 2022-23
Assessment Framework
COMPUTER SCIENCE Grade-IX (SSC-I)
Details of Content Areas/ SLOs

Content Domain/ Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
A: Computer Systems	[SLO CS-09-A-01] Students will define and describe types of systems (artificial, natural), computer hardware components such as computer architecture (CPU, microprocessors, etc.) Student will understand that • Brief history of computer systems and generation of computers • Basic concept of a system, types of systems • Name and identify the core components of a computer (input/output devices, system unit(motherboard, memory, CPU, power supply, etc.), and data storage devices • Understand and identify Von Neumann Architecture • Computer architecture How data is transmitted within a computer system • What are some of the requirements for a functioning computer systems and what are some key concepts • The core parts of a computer system and how they all work together, including definitions and key functions of computer architecture such as central processing unit (CPU), arithmetic logic unit (ALU), control unit (CU), memory, operating system and application software, and data representation in computers (bit, byte, binary, denary/ decimal, hexadecimal) • Difference between hardware engineering and software engineering	Summative for Theory and Practical Based Assessment	Knowledge/ Understanding	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	40

Content Domain/ Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
	• Difference between natural and artificial systems • Types and hierarchy of memory with respect to their volatility/retention, speed, storage capacity, cost • Difference between necessary and auxiliary components of a computer system Students will be able to • Recognize and describe key components of a computer system, such as: ○ Differentiate between natural and artificial systems ○ Outline the architecture of the central processing unit (CPU) and the functions of the arithmetic logic unit (ALU) and the control unit(CU) and the registers within the CPU. ○ Differentiate between primary memory, cache, and secondary memory. ○ Describe the main functions of operating systems and application systems. • Explain how the various components interact together to transmit data and instructions • Illustrate the hierarchy of memory and storage devices with respect to their volatility/retention, speed, storage capacity, cost • Differentiate between necessary and auxiliary components of a computer system				
	[SLO CS-09-A-02] Students will be able to identify and explain system software, application software, low-level and high-level programming languages, and their uses. Students will understand that • The main functions of systems software with some examples	Summative for Theory and Practical Based Assessment	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	

Content Domain/ Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
	- The main functions of operating systems and application software - Outline the uses of various application software. Students will be able to - Select an appropriate medium to create artifacts (Planning the document / information flow, editing and alignment of page, paragraphs, text, tables, and graphics) to communicate ideas in various digital tools such as: - Image processing tools (like Photoshop, Canva.com, GIMP etc.) - Word processors (like MS Word, Google Docs etc.) - Presentations (like MS PowerPoint or Google Slides etc.) - Spreadsheets (like MS Excel or Google Sheets)				
	[SLO CS-09-A-03] Students will be able to identify and analyze data communication, computer networks, networking devices, basic networking systems and understand how data is transmitted and key concepts such as protocols, speeds, etc. Students will understand - How data is transmitted across a computer network for example circuit switching, packet switching, layering, encapsulation, and protocols - Common network topologies and transmission modes - Outline the advantages and disadvantages of wireless networks - Outline the advantages and evolution of the Internet - Outline common applications of the Internet - The internet is the largest computer network ever built and learn how it works	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	Identify common network problems and provide possible solutions				
B: Computational Thinking and Algorithms	[SLO CS-09-B-01] Understand and apply techniques to decompose problems. [SLO CS-09-B-02] Solve simple and complex problems computationally. Students will understand - The importance of computational thinking and problem-solving in computer science. - Principles of computational thinking: - Logical thinking - Algorithmic thinking - How to identify steps in identifying a computing problem - How to identify the inputs, processes, and outputs of a problem - Different methods to design and construct a solution to a simple problem, such as flow charts, and/or concept maps - Steps to produce simple diagrams to show: - The structure of a problem - Subsections and their links to other subsections Students will be able to - Explain the role of computational thinking in computer science - Read and interpret simple computational problems - Apply computational thinking principles to define and refine problems - Identify the procedure appropriate to solving a problem - Evaluate whether the order in which activities are undertaken will result in the required outcome. - Identify the inputs and outputs required in a solution	Summative for Theory and Practical Based Assessment	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	25

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C: Program ming Fundamentals	[SLO CS-09-C-01] Students will understand web development and differentiate between a website and a web application. Students will - Grasp the basic concepts and principles of web development, including the technologies, languages, and frameworks commonly used in building websites and web applications. - Understand the distinctive features of web applications. They should learn that web applications are dynamic in nature, offering interactive functionalities, data processing, user authentication, and real-time updates. - Comprehend that websites primarily provide information and content to users, while web applications offer more complex functionality, such as user input, data manipulation, and task execution. Students will be able to... - Distinguish between front end and back-end development - Determine which technology is appropriate for making static and dynamic web pages and web applications - Identify and describe the features and characteristics of a website. This may include static content, informational pages, navigation menus, multimedia elements, and contact forms. - Distinguish between websites and web applications based on specific criteria.	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	40
	[SLO CS-09-C-02] Students should be able to create a static website using HTML/CSS in an appropriate environment. Students will know What is HTML	Summative for Theory and	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	

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	- What is CSS - What is JavaScript Students will understand - Characteristics, examples, similarities & differences between of static and dynamic website - The purpose of HTML i.e. to display text and image content over the Internet - The capability of HTML documents to hyperlink to other documents/pages - The structure of an HTML document including the underlying DOM tree (Document Object Model) - How JavaScript is used to modify a website Students will be able to - Differentiate between a static and dynamic website - Create a static website using HTML & CSS in an IDE like Visual Studio, NetBeans etc.	Practical Based Assessment			
	[SLO CS-09-C-03] Students should be able to create dynamic websites using JavaScript as the frontend scripting. Students will understand - Variables, Data Types and Arrays in JavaScript - Variable assignment in JavaScript - What aspects of HTML can be changed with JavaScript - What aspects of CSS can be changed with JavaScript Students will be able to - Use JavaScript to modify an HTML website to create a dynamic website	Summative for Theory and Practical Based Assessment	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	

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	[SLO CS-09-C-04] Students should be able to implement common algorithms that use sequence, selection, and repetition in JavaScript. Students will understand • An algorithm is a set of instructions • How the sequence of instructions affects the result • Sequence, Selection and Repetition in JavaScript • Arrays in JavaScript • Students will be able to • Construct an array and populate its values using a loop in JavaScript	Summative for Theory and Practical Based Assessment	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	
	[SLO CS-09-C-05] Students will determine ways of debugging their code in JavaScript Students will understand • Code written outside of a function is hard to test • Code written inside a function can be tested • That they can write code that calls functions to ensure the results are correct • Using a debugger allows programmers to set a breakpoint to stop execution of their code to see the state of variables mid-execution for the purpose of discovering errors in their code Students will be able to • Write code to invoke functions and check their return values for correctness • Set a breakpoint to debug mistakes in their code	Summative for Theory and Practical Based Assessment	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	
D: Data and Analysis	[SLO CS-09-D-01] Students will explain the scope of the data science field as an interdisciplinary field (computer sciences, mathematics & statistics, and business knowledge & understanding).	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	25

Content Domain/ Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
	- Students will understand the key concepts and principles of data science - Students will know definitions of key terms: data science, data analytics, computer science, mathematics, statistics, and business knowledge & understanding Students will be able to: - Identify the difference between data science, data analytics, computer science, mathematics, statistics, and business knowledge & understanding - Students will develop analytical thinking skills by reading case studies on data science applications - Read a case study or a newspaper article or online paper to identify a problem that can be solved using data science				
	[SLO CS-09-D-02] Students will define and explain data types, data collection, and data storage. Students will understand: - Data is observations or measurements represented as numbers, text, or multimedia(image, sound, video, etc.)and the dataset is a collection of this data that is related in some context. - Different sources of data like sensors - The concept of dataset - Table (relation) - Rows (record, tuple, object) - Column (object attributes, properties) - Types Object attributes or properties: - Categorical or Qualitative Attribute - Nominal	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	○ Binary ○ Ordinal ● Numeric or Quantitative ○ Interval-scaled, ○ Ratio-scaled, ○ Discrete ○ Continuous ● Data collection, analysis, and visualization. ● Students will know how to use summary statistics to formulate sentences and describe the data, data collection methodology, how the data will be used Students will be able to: ● Explain how data, information, and knowledge are represented for computational use. ● Collect, upload, and share personal data collected in class(e.g., Stick Figure exercise in Learning Activity#2) ● Learn about different representations of distributions using software. ● Utilize software to begin to analyze plots of data collected ● Interpret different types of data: Numerical and graphical summaries. ● Understand that rows and columns area form of data structure. ● Explain why the relationship between the variables might exist, or, if there is no relationship, why that might be so ● Construct and interpret a frequency table. ● (Advanced) Load data into RStudio. Students will be able to: ● Distinguish between data and dataset ● Collect data from different sources				

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	- Identify missing data in a dataset - Construct summary statements about the data, how it is collected, how it is used, and how to work with it - Explain why the relationship between the variables might exist, or if there is no relationship, why that might be so. (Advanced) Construct and interpret a frequency table (Advanced) Understand the connection of databases to machine learning. (Advanced) Understand the differences between structured and unstructured data, quantitative and qualitative data)				
	[SLO CS-09-D-03] Students will be able to define and explain big data, and applications of big data in real-world business Students will understand - What insights big data can provide to businesses - What tools and systems are used by big data scientists and engineers - How predictive modeling and graph analytics can be leveraged to model problems? (Advanced) Be able to ask the right questions about data and do basic exploration of large, complex datasets. - Students will know... - Applications of bigdata - What makes big data valuable? - Characteristics of big data - How data science gets value out of big data - Introduction to concepts such as the Internet of Things(IoT) and the role of the cloud. Students will be able to	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	- Identify the different types of big data - Recognize if big data is used to solve a business problem - Read and critique published stories by interpreting the visualizations and experimental conclusions - Formulate questions, identify existing or similar business cases, and evaluate how the new business solutions that leverage big data stack up against the old.				
E: Applications of Computer Science	[SLO CS-09-E-01] Students will be able to describe uses and applications of computing like AI, Machine Learning, and Cloud Computing Students will understand... - What is AI and Machine learning and the kind of problems each is able to solve - Different areas of AI–Speech recognition, Computer Vision, Natural Language Processing, and Expert Systems. - Different applications of AI in domains like health care, education, gaming, agriculture. - What is cloud computing and the need for it Students will be able to... - Identify applications that can be solved using these technologies and techniques - Identify different applications of AI	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	10
	[SLO CS-09-E-02] Students will be able to discuss the social implication of the usage of AI in decision making that affects humans Students will understand...	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	- AI algorithms make decisions that work for most people but harm or disadvantage others. - Designers to AI algorithms should keep their focus on benefiting people Students will know... - Ethical issues in some popular AI tools Students will be able to... - Explain what some ethical issues are when computers make decisions for humans - Identify decisions that might harm humanity - Identify decisions that might be biased towards are certain group of individuals				
F: Impacts of Computing	[SLO CS-09-F-01] Understand and apply safe and responsible use of computers (responsible use of hardware, appropriate use of software, and safe use of digital platforms like data searches, social networking, etc.) Students will understand... - The selection and use different hardware, software, and digital platforms for safe use depending on the tasks to be executed - The need for laws to protect user privacy and intellectual property Students will know... - What is meant by the key concepts of data ethics and intellectual property rights - How to safely use data searches and social networking sites - Legal issues related to computing such as software piracy laws Students will be able to...	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	15

Content Domain/ Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
	- Discuss how ethics are guidelines that dictate the responsible use of computing systems - Identify computer related laws - Interpret software agreements, licenses, and application terms				
	[SLO CS-09-F-02] Analyse the beneficial and harmful effects of computing innovations such as social networking, fake news, etc. Students will understand... - Every computing innovation will solve a need but can cause harm to users as well - Benefits can be the need being solved and the societal & user benefits - Harmful effects of computing innovations on users, society, environment, etc. - The tradeoffs between information privacy, system security and usability - The importance of designing computing systems that will protect user privacy and increase system security Students will know... - What is meant by key concepts and be able to differentiate between them: scams, software piracy, freeware, shareware, opensource, malware, phishing, hacking - The tensions between censorship of internet and freedom of speech - Common types of security problems such as spam, spyware, pharming, cookies etc. Students will be able to... - Explain basics of reliable and unreliable sources of information	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	- Distinguish between fake news and credible sources of information - Identify malicious internet scams, phishing, pharming, fraudulent activity, fake websites - Evaluate designs of computing technologies(e.g. social media, generative AI etc.) based on their user privacy policies and agreements				
	[SLO CS-09-F-03] Evaluate the ways computing impacts personal, ethical, social, economic, and cultural practices. Students will understand... - Environmental impacts of computing - Computing has changed the way we live and conduct business - Impact of computing on globalization and e-commerce Students will know... - The key terms: patents, trademarks, copyrights Students will be able to... - Strategize on how to minimize the environmental impacts of computing - Search and identify patents, trademarks, and copyright information for computing applications - Determine if technology design maybe infringing on intellectual property - Evaluate designs of computing technologies in terms of personal, environmental, ethical, legal, social, economic, and cultural impacts	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	
G: Digital Literacy	-	-	-	Not applicable for grade 9.	-

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H: Entrepreneurship in the digital age	[SLO EN-09-H-01]: Students identify a problem and create a business using design thinking Students will understand: • Skills needed for problem solving: 4Cs of the 21st century (collaboration, critical thinking, creativity, communication) • Define design thinking as a process which includes: ○ Empathizing to discover the problem ○ Defining or interpreting the problem ○ Ideating on how to solve the problem ○ Experimenting or prototyping the solution ○ Evolving the solution as insights redefine the problem • How to define & articulate an entrepreneurial solution to address a local or global problem • Sustainable development goals(SDGs) as the world's development goals, and linking these SDGs to potential problems that need resolving • Growth mindset vs. Fixed mindset Students will be able to: • Define a problem in their local context • Categorize the problem according to SDGs • Use design thinking to create a solution • Identify advantages of having a growth mindset in entrepreneurship	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	10
	[SLO EN-09-H-02]: Students will use digital tools to create and present a business plan for an entrepreneurial solution. Students will understand: • Key elements of a business plan containing at least the following	Summative for Theory and	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as Lab work will be assessed in the Practical Based Assessment	

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	○ Market size(in terms of TAM total addressable market, SAM Serviceable Available Market, and SOM Serviceable Obtainable Market) ○ Solution product/service & unique selling proposition ○ Financial feasibility ○ Go to market strategy ● How to collect & synthesize information to create a business plan ● How to evaluate a business plan Students will be able to: ● Create a business plan using digital tools like MS Word or Google Docs to communicate their business problem, suggested solution, and financial feasibility ● Evaluate a business plan using criteria such as communication, feasibility, market potential etc.	Practical Based Assessment			